FastAPI baseline: Locust load test results

Arabic sentiment service, before optimization. Data extracted from the four Locust HTML reports of 25 Oct 2026.

Setup

Service: FastAPI + uvicorn (single worker), PyTorch FP32 on an RTX 4050 (WSL). Locust and the API run on the same laptop. Each simulated user waits 0.5 to 2.0 s between requests and sends a random real review from `data/processed/test.csv` (90% /predict, 10% /health). Stats were reset after ramp-up; each run lasts about 3 minutes. Numbers below are the **POST /predict** row.

Results

Users	Run start (UTC)	Duration	/predict requests	Fails	p50	p95	p99	Avg latency	Avg RPS /predict (all)
1	2026-10-25 12:27:21	3 minutes and 10 seconds	122	0	78 ms	140 ms	160 ms	85 ms	0.6 (0.7)
50	2026-10-25 12:31:46	3 minutes and 9 seconds	5,453	0	180 ms	530 ms	1100 ms	215 ms	28.8 (31.6)
100	2026-10-25 16:48:26	3 minutes and 4 seconds	5,955	0	1400 ms	1900 ms	4200 ms	1434 ms	32.3 (35.6)
1000	2026-10-25 17:00:15	3 minutes and 26 seconds	6,721	0	26.0 s	29.0 s	29.0 s	26.0 s	32.6 (35.9)

What the runs show

1. Throughput plateaus at about 32 predict RPS (36 total). From 50 users upward RPS stays flat while latency grows, so extra users only add queueing. Going from 100 to 1000 users added about 0.2 RPS and made p50 latency about 18 times larger.

2. Latency follows the queueing law latency = users / RPS - 1.25 s (think time): 100 users predicts 1.6 s (observed average 1.4 s); 1000 users predicts 26.6 s (observed 26.0 s).

3. The saturation point is about 48 users = max RPS x (single-user latency + think time) = 36 x (0.085 + 1.25). 50 users sits on the knee (noisy p95); 100 users is a stable saturated state.

4. 0 failures does not mean healthy. Locust has no request timeout by default, so at 1000 users every request succeeded but took about 26 s; a real client with a 30 s timeout would be failing.

5. Single-user cost is 85 ms per real review (p95 140 ms), about 4x the cost of short hand-written sentences, because real reviews are longer.

Throughput and latency over time

Left: requests per second (blue) and active users (grey). Right: p50 and p95 latency. Time starts at the stats Reset.















